

NCERT Practical Assignment #01

Reflection Report

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02 July 2026

Introduction

This report reflects on my experience completing Practical Assignment #01, which required installing Ubuntu Linux, deploying the Nginx web server, and developing and hosting a professional front-end website for the National Cyber Emergency Response Team (NCERT) at <http://localhost:8080>.

What I Learned

Through this assignment, I gained hands-on experience with several core skills:

- Installing and configuring the latest stable version of **Ubuntu Linux** on a virtual machine, including applying system updates.
- Installing and verifying the **Nginx** web server, including checking its running status and version.
- Configuring an Nginx server block to serve a static website on a custom port (8080) instead of the default port 80.
- Structuring and building a **responsive front-end website** using HTML5, CSS3, and JavaScript, without relying on any backend.
- Understanding the major operational domains of a real-world Cyber Emergency Response Team, namely Cyber Security, the Security Operations Center (SOC), Software Development, Governance/Risk/Compliance (GRC), and Digital Forensics – and how these teams work together through a structured incident response lifecycle (detection, alerting, analysis, containment, eradication, recovery, reporting, and lessons learned).
- Working with modern development tools and workflows to design a visually polished, professional user interface under a tight deadline.

Challenges Faced

1. Storage Issue During Ubuntu Installation

While installing Ubuntu on my virtual machine, I ran out of disk space partway through the installation, which caused the installer to fail.

2. Risk of Losing Time While Fixing the VM

Since the Ubuntu installation issue needed to be fixed before I could deploy anything with Nginx, I was at risk of falling behind schedule for the rest of the assignment.

3. Achieving a Professional, Aesthetic Design Quickly

Designing a website that looked professional, modern, and visually appealing – while also being fully responsive – was challenging to do well within the limited time of a single lab session.

How I Resolved These Challenges

- **Storage issue:** I identified that my host machine did not have enough free disk space allocated to the virtual machine. I resolved this by clearing up unnecessary files and freeing additional storage space on my system, then resizing/reallocating the virtual disk so the Ubuntu installation could complete successfully.
- **Working in parallel:** To avoid stopping my progress while the VM issue was being resolved, I switched to working on the website's front-end code in **VS Code on Windows** in parallel. This let me continue building and testing the HTML, CSS, and JavaScript for the NCERT website locally, so that once the Ubuntu environment was ready, I only needed to copy the finished project over and deploy it with Nginx – rather than losing time developing everything from scratch on the VM.
- **Design and speed:** To achieve a more polished and professional look within the time available, I used AI assistance while building the website's design and layout. Since this is the AI era, I believe it is both practical and expected to use AI tools to improve design quality and speed up development – it helped me iterate on a clean, modern, responsive layout much faster than I could have manually.

Conclusion

Overall, this assignment strengthened my practical understanding of Linux system administration, web server deployment, and front-end development, while also giving me insight into how a national-level Cyber Emergency Response Team is organized and how it responds to security incidents. The storage issue taught me the importance of planning adequate resources before starting a VM-based installation, and working in parallel across environments helped me manage my time effectively despite the setback.